



The Silent Asset

Franchise Deposits and Cross-Sectional
Bank Equity Returns in the Euro Area



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MSc Dissertation Defense 2026 | Supervised by Prof. Carla Soares

Key Terminology

A decade of silence, and then a regime change...



MATURITY TRANSFORMATION

When long-term rates are higher than short-term rates, banks can earn more on their assets while funding costs (such as for deposits) remain relatively low.

$$\begin{aligned} \text{NIM} &= \frac{\text{Net Interest Income}}{\text{Average Interest Earning Assets}} \\ &= \frac{\text{Interest Income} - \text{Interest Expense}}{\text{Average Interest Earning Assets}} \end{aligned}$$

DEPOSIT FRANCHISE

Earnings generated by a bank's ability to retain customer deposits at low, stable rate below the rest of the market while reinvesting into assets.

Franchise-like deposits can be retained at low and stable rates relative to market.

DEMAND DEPOSITS

Transactional, liquid deposits that can typically be withdrawn on demand and tend to reprice slowly.

SAVINGS DEPOSITS

Interest-bearing deposits that are more yield-sensitive and more likely to reprice or migrate when rates rise.

Motivation

A decade of silence, and then a regime change...



From 2011 to 2021, low and negative policy rates compressed the value of cheap deposit funding. The ECB's 2022-24 hiking cycle reversed that: policy rates rose from below zero to 4%, the highest since the financial crisis. This may have made deposit repricing economically relevant again.

<0% → 4%

ECB deposit facility rate,
2022-24 hiking cycle

BANK A: FRANCHISE-FUNDED

Sticky deposits reprice slowly even when rates rise. When the curve steepens, asset yields rise while funding costs lag, supporting bank profitability.

BANK B: NON-FRANCHISE-FUNDED

Non-franchise depositors want returns. Deposits reprice fast or migrate to better-paid products – the same steepening may hurt these banks.

Do equity markets price this reinvigorated advantage?

Research Question

Euro-Area + Equity Market + Yield Curve Movements + Other Conditionalities



H1: Is franchise-deposit exposure asymmetrically priced in the euro-area equity market across episodes of yield-curve steepening and flattening?

...and is this pricing conditioned by:

H2: DEPOSIT TYPE

Are demand deposits treated better than savings deposits during steepening?

H3, H4: MARKET REGIME

Do rate level and systemic stress condition franchise deposit value?

H5: ASSET-SIDE DEPLOYMENT

Do deposits and assets **jointly** create an advantage?

Yes, with conditional support.

H1: Qualified

Deposit exposure is priced asymmetrically, with a relative premium for high-customer-deposit banks during steepening.

H2: Strong

Markets differentiate between demand and savings deposits. Differential driven by a penalty on Savings-heavy banks during curve steepening.

H3: Supported

Demand and Savings deposits are treated differently post-2022 hiking cycle, consistent with franchise-deposit value re-emerging when the rate regime becomes supportive.

H4: Supported

High-systemic-stress months reveal a stronger penalty on savings-heavy banks.

H5: Supported

The steepening premium is concentrated amongst banks that combine high deposit and loan exposure.

The Literature

The literature identifies five mechanisms but rarely tests them jointly in the euro-area equity market.



H1

NON-LINEAR LINK CURVE-PROFITABILITY

Alessandri & Nelson (2015) yield curve level and slope lift long-run profits, even after accounting for trading
Borio et al. (2017) slope-profit link is positive but concave; steepening benefit larger when curve is flat or inverted.
Gambacorta & Iannotti (2007) bank-rate pass-through differs after tightening vs easing shocks

H2

DEPOSIT TYPES MATTER

Kok Sørensen & Werner (2006) pass-through demand 0.15 vs savings 0.84 in euro-area; demand are much stickier.
De Bondt (2002) overnight deposits show low pass-through.
Gropp et al. (2007) pass-through is sluggish for demand (and savings) deposits.

H3

RATE LEVEL CONDITIONALITY

Claessens et al. (2018) low rates compress NIM further
ECB (2015) persistent lows erode smaller banks, especially for smaller banks less able to hedge interest-rate risk.
Heider et al. (2019) Banks resisted passing negative rates to depositors during 2014; higher funding costs.
Pancaro et al. (2025) deposit franchise moved from liability to positive asset post-rate normalization.

H4

SYSTEMIC STRESS CONDITIONALITY

Berlin & Mester (1999) core funding worth more in fragility
Cornett et al. (2011) core-funded banks lent relatively more in 2008.
Acharya & Mora (2015) crisis deposit inflows can weaken

H5

ASSETS & DEPOSITS MATTER JOINTLY

Ippolito et al. (2018) floating-rate loans make asset cash flows more rate-sensitive.
Egan et al. (2022) deposit-taking and lending are synergistic. Deposit productivity is large portion of value.
Ho & Saunders '81; Angbazo '97 bank margins depend on both sides of the balance sheet.

Literature Gap

Existing evidence explains the channels separately, but not their joint pricing in euro-area bank equities.



ACCOUNTING PROFITABILITY, NOT CROSS-SECTIONAL EQUITY PRICING

Slope and level support accounting profitability, non-linearly and regime-dependently. We test these conditionalities in the equity market.

Alessandri & Nelson 2015 · Borio et al. 2017

U.S. BASED FRANCHISE DEPOSIT VALUE LITERATURE

Sticky, low-beta deposit value is largely US, accounting-side evidence. This thesis studies the equity reactions in the euro-area.

Drechsler et al. 2017, 2021 · Egan et al. 2022

EQUITY-BASED LITERATURE: FOCUSES ON MONETARY POLICY ANNOUNCEMENTS

Equity-based literature uses event studies around monetary policy announcements. This thesis studies equity reactions.

Flannery & James 1984 · English et al. 2018 · Ampudia & Van den Heuvel 2022

THE GAP

Channels are studied in isolation; steepening and flattening are treated as symmetric shocks; and deposit composition is less often tested in equity-market terms. This thesis joins all three in a single euro-area cross-sectional design.

Data & Sample

Monthly panel of listed euro-area banks, April 2011 - December 2024



165

months,
April 2011 - Dec 2024

36 / 5,318

Banks / observations
in the **main sample**

31 / 3,386

banks / observations in the
**deposit composition
subsample**

YIELD CURVE VARIATION

Steepening months: **2,618 obs.** · avg. non-zero move: **+8.6 bps**
Flattening months: **2,700 obs.** · avg. non-zero move: **−9.7 bps**

SOURCES

Refinitiv Datastream / Worldscope · BvD OSIRIS ·
ECB Data Portal · Wu-Xia shadow rate

SAMPLE LIMITATIONS

Slow-Moving Accounting Variables may miss within-year balance sheet fluctuations

- Interacted deposit exposure with steepening to create temporal cross-sectional variation.
- Bank clustered SEs + WCB p-values

Euro-Area Listed-Bank Scope

Country-specific variation may affect bank returns

- Include sovereign spread and systemic risk controls
- Country FE may be a nice future extension.

Customer Deposits are a broad proxy

- Decompose into demand and savings components.
- Re-run H1 CustDep tests on Composition sample.

Identification Design

Cross-sectional reactions to yield-curve movements, paired with strict inference design



DEPENDENT: EXCESS EQUITY RETURNS

- Provide forward looking insight that current profitability may not capture.
- Returns are noisy (English et. Al, 2018; Bernanke and Kuttner, 2005)

BANK + MONTH FE DESIGN

Month FE are necessary to address English/Bernanke criticisms.

*Month FEs remove common monthly effects.
Bank FEs absorb persistent bank traits.*

$$\begin{aligned} r_{it}^{\text{excess}} &= \alpha_i + \lambda_t \\ &+ \beta_S (\text{Steepening}_t \times \text{HighDepExposure}_{i,t-12}) \\ &+ \beta_F (\text{Flattening}_t \times \text{HigDepExposure}_{i,t-12}) \\ &+ \dots + \varepsilon_{it} \\ \alpha_i &= \text{Bank FE}, \lambda_t = \text{Month FE} \end{aligned}$$

β_S/β_F : How much excess returns do higher deposit exposure banks earn each month when the yield curve steepens/flattens?

FOCUS ON CROSS-SECTIONAL DIFFERENCES

Hoffman et. Al (2019) and others show that cross-bank heterogeneity is substantial.

Banks are split into high and low deposit exposure via yearly lagged median.

ASYMMETRIC DECOMPOSITION

$$\text{Steepening}_t = \max(\Delta \text{slope}_t, 0)$$

$$\text{Flattening}_t = \min(\Delta \text{slope}_t, 0)$$

$$\Delta \text{slope}_t = \Delta 10Y-2Y OIS$$

HIGHLY CONSERVATIVE INFERENCE

Bank-clustered standard errors + Wild-cluster bootstrapped p-values (999 Rademacher draws)

Cluster count is low **G=36/31** → MacKinnon and Webb (2017), Cameron et al. (2008) show clustered SEs can over-reject with few/uneven clusters → WCB p-values provide stricter inference.

Result Stack



H1

Equity markets price banks with stronger franchise deposit exposure more favourably during yield-curve steepening than flattening.

$$\beta_S: Steepening_t \times HighCustDep_{i,t-12}$$
$$\beta_F: Flattening_t \times HighCustDep_{i,t-12}$$
$$H_0: \beta_S - \beta_F = 0$$

H2

Banks with higher demand-deposit shares are priced more favorably than banks with higher savings-deposit shares during yield-curve steepening.

$$\delta_D: Steepening_t \times HighDemand_{i,t-12}$$
$$\delta_T: Steepening_t \times HighSavings_{i,t-12}$$
$$H_0: \delta_D - \delta_T = 0$$

Baseline Controls

$$X_{i,t-12} = [Tier1_{i,t-12}, Size_{i,t-12}]$$

Expanded Controls

$$ROAA_{i,t-12}$$
$$CreditRisk_{i,t-12}$$

Macro Controls

$$\Delta Short_t \times DepExp_{i,t-12}$$
$$\Delta CISS_t \times DepExp_{i,t-12}$$
$$\Delta SovSpread_{c(i),t}$$

H3

60-Month Rolling Window,
Quarterly Increments

H4

Split sample into
High and Low CISS Median
and re-run H1/H2

H5

Deposit x Loan Quadrants

$$\theta_{HH}(Steepening_t \times HighDepHighLoan_{i,t-12})$$
$$\theta_{HL}(Steepening_t \times HighDepLowLoan_{i,t-12})$$
$$\theta_{LH}(Steepening_t \times LowDepHighLoan_{i,t-12})$$

Full Robustness Stack

Controls → Alternative Slopes → Winsorised → Leave-One-Bank/-Country-Out → Continuous Specifications

Winsorisation

Winsorising dependent variable, monthly excess and macro indicators at the 1st and 99th percentiles. Outlier Check.

Leave-One-Bank / -Country-Out

Remove one bank and country at a time and re-estimate

Alternative Slopes

5Y - 2Y, 10Y - Shadow/DFR

Continuous Specifications

Replace median exposure splits with standardized z-score deposit terms. Checks that results are NOT driven by binary split of sample.

Common Sample Bridge

Re-run H1 on the demand decomposition sample.



Results



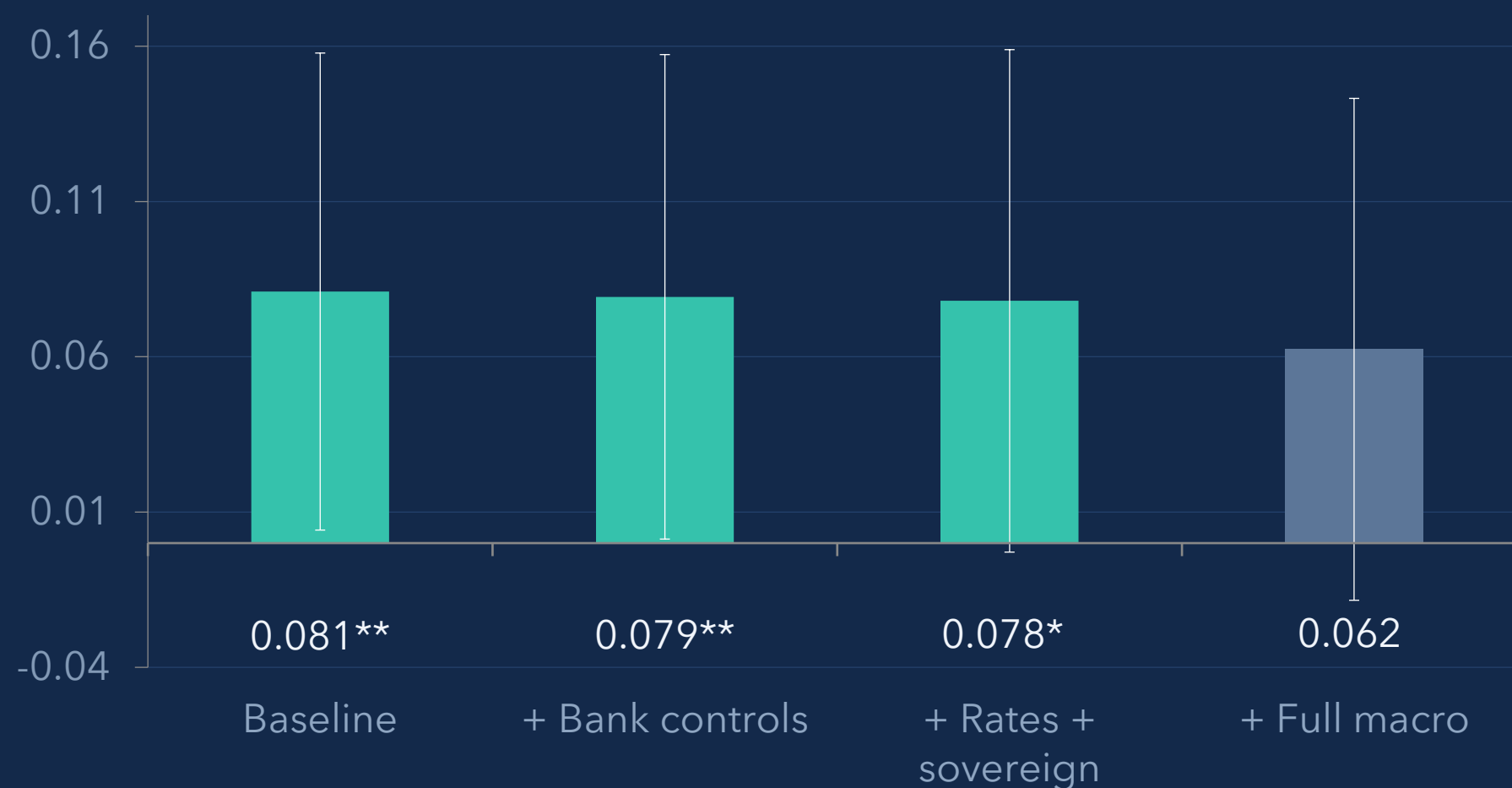
H1

Equity markets price banks with stronger franchise deposit exposure more favorably during yield-curve steepening than flattening.

$$\beta_S(\text{Steepening}_t \times \text{HighCustDep}_{i,t-12})$$
$$\beta_F(\text{Flattening}_t \times \text{HighCustDep}_{i,t-12})$$
$$H_0: \beta_S - \beta_F = 0$$

Broad deposit exposure gives the initial signal.

β_S coefficient across specifications



Final bar loses significance under full macro/stress controls ($p = 0.130$)

+0.81 pp (WCB $p = 0.050^{**}$)

monthly excess return per 10 bp of steepening, high- vs low-deposit banks

- β_F Flattening response (**n.s.**)
- $\beta_S - \beta_F \approx +0.99$ per 10 bp (WCB $p = 0.078^*$)
- Indicates a steepening premium, but the effect attenuates once heavier macro-financial interactions are added.

H1: QUALIFIED SUPPORT

H2

Banks with higher demand-deposit shares are priced more favorably than banks with higher savings-deposit shares during yield-curve steepening.

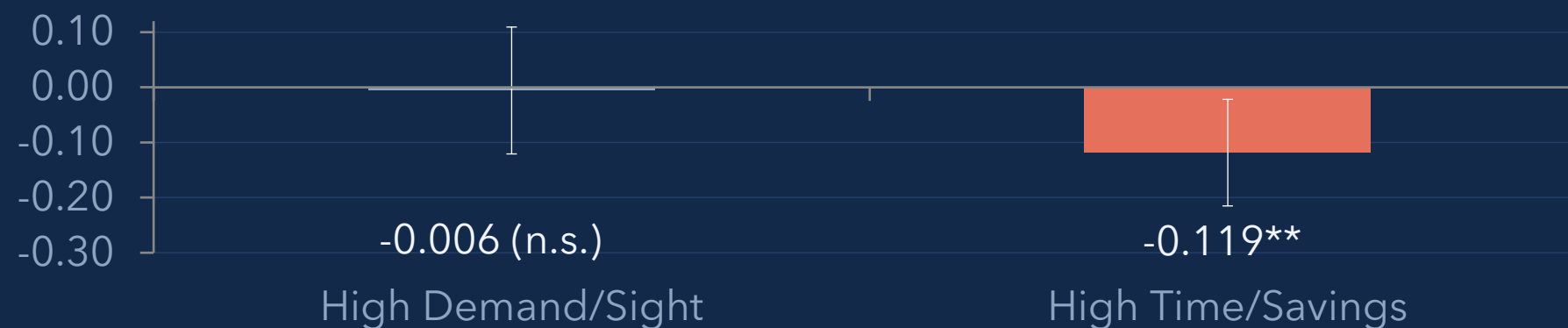
$$\delta_D(\text{Steepening}_t \times \text{HighDemand}_{i,t-12})$$

$$\delta_T(\text{Steepening}_t \times \text{HighSavings}_{i,t-12})$$

$$H_0: \delta_D - \delta_T = 0$$

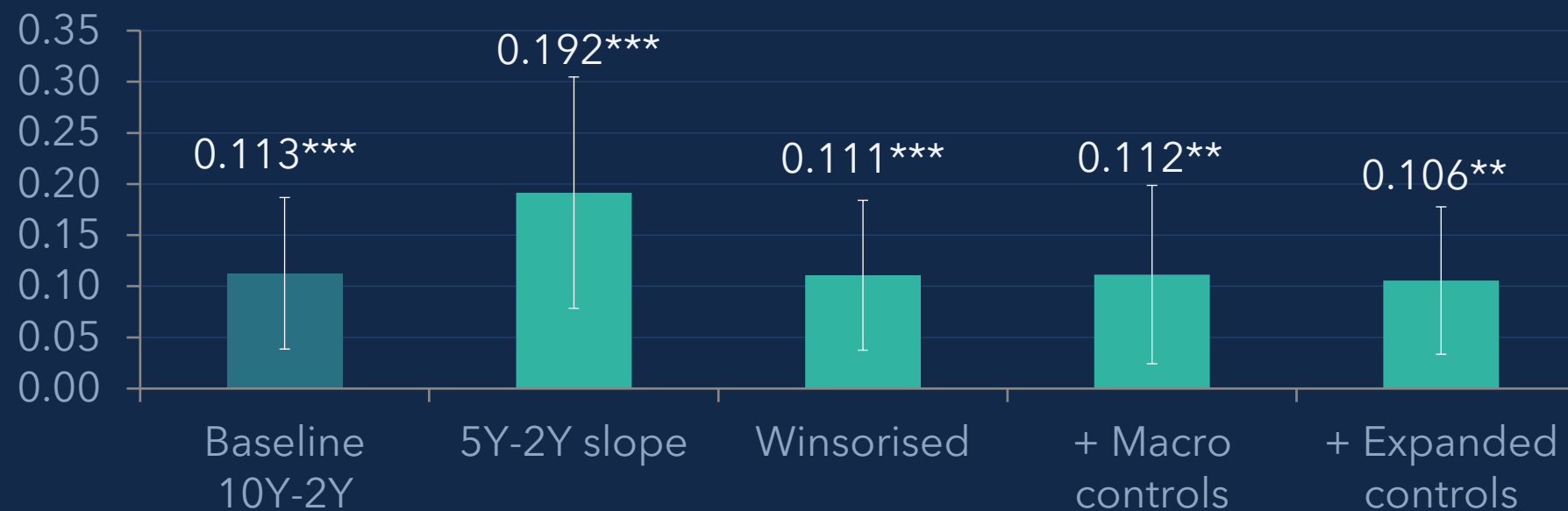
The market prices based on deposit type.

δ_D and δ_T Baseline Model



Savings-heavy penalty of 1.19 pp per 10 bp of steepening drives differential.

$\delta_D - \delta_T$ steepening differential survives robustness



+1.13 pp (WCB p = 0.007)

Demand – Savings differential per 10 bp of steepening

- WCB p ≤ 0.024 in most robustness tests

What does not survive?

- Continuous z-scores: Steepening × DemandZ is positive (+0.068, WCB p = 0.031), but this answers a marginal-intensity question, not the main high-vs-low group question.
- 10Y-shadow/DFR slope: markets price the observable curve.

H2: STRONG SUPPORT

H3

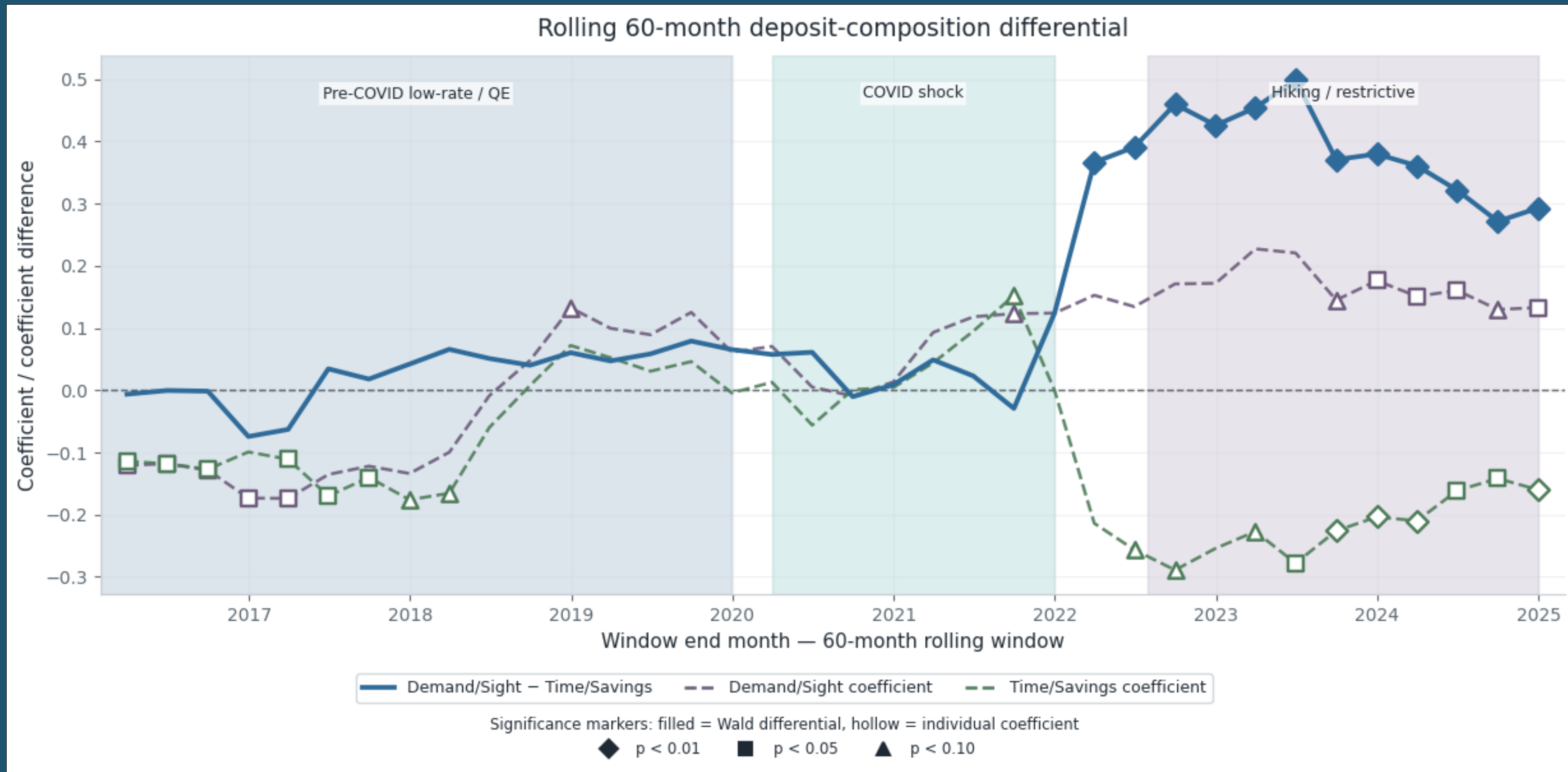
The value of franchise deposit exposure during steepening is weaker in low-rate regimes and reappears in high-rate regimes.

60-month Rolling Windows in Quarterly Increments

$$\delta_D(\text{Steepening}_t \times \text{HighDemand}_{i,t-12})$$

$$\delta_T(\text{Steepening}_t \times \text{HighSavings}_{i,t-12})$$

$$H_0: \delta_D - \delta_T = 0$$



H3

The value of franchise deposit exposure during steepening is weaker in low-rate regimes and reappears in high-rate regimes.

60-month Rolling Windows in Quarterly Increments

$$\delta_D(\text{Steepening}_t \times \text{HighDemand}_{i,t-12})$$

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$$H_0: \delta_D - \delta_T = 0$$

Silent through QE, visible after 2022

2011 – 2019

LOW RATE / QE

Differential ≈ 0 , n.s.

2020 – 2021

COVID SHOCK

No Clean Signal.

2022 – 2024

HIKING

Demand – Savings gap turns large, positive and significant in rolling windows.

The market starts separating transactional balances from yield-chasing balances exactly when rates are high and franchise value can matter again.

H3: SUPPORTED

H4

Franchise deposit exposure is priced conditional on the stress environment.

$$\begin{aligned} &\delta_D(\text{Steepening}_t \times \text{HighDemand}_{i,t-12}) \\ &\delta_T(\text{Steepening}_t \times \text{HighSavings}_{i,t-12}) \\ &H_0: \delta_D - \delta_T = 0 \end{aligned}$$

Savings-deposits are punished in stress

LOW-STRESS MONTHS (CISS ↓)

+1.21 pp (WCB $p = 0.049$)

Differential per 10 bp steepening

Neither individual coefficient significant alone.

Investors look at growth, capital, efficiency.

HIGH-STRESS MONTHS (CISS ↑)

−1.82 pp (WCB $p = 0.047$)

Penalty for savings-heavy banks per 10 bp steepening

+1.03 pp Differential per 10 bp steepening
($p = 0.08$)

Unstable funding is punished when market funding is fragile.

H4: SUPPORTED

H5

Asset repricing and deposit exposure jointly shape the favorable valuation of banks during steepening episodes.

$$\begin{aligned} &\theta_{HH}(\text{Steepening}_t \times \text{HighDepHighLoan}_{i,t-12}) \\ &\theta_{HL}(\text{Steepening}_t \times \text{HighDepLowLoan}_{i,t-12}) \\ &\theta_{LH}(\text{Steepening}_t \times \text{LowDepHighLoan}_{i,t-12}) \end{aligned}$$

Premium concentrated in banks that are heavy in BOTH loans AND deposits.

HIGH DEP / LOW LOAN

−0.06 pp

n.s.

HIGH DEP / HIGH LOAN

+1.01 pp

per 10 bp · WCB p = 0.034

LOW DEP / LOW LOAN

Reference

omitted category

LOW DEP / HIGH LOAN

+0.03 pp

n.s. – assets without the funding edge

- The premium lives in the combination. Not deposits alone. Not loans alone.
- Loan share proxies the asset-side repricing channel:
 - Large loan book banks are more exposed to interest-rate sensitivity on asset-side of the balance sheet (Flannery et al., 1984).
- **COARSE:**

Wald Tests

- $\theta_{HH} - \theta_{HL} = +1.07 \text{ pp}^{**}$
- $\theta_{HH} - \theta_{LH} = +0.98 \text{ pp}^*$

H5: SUPPORTED

Limitations and Future Extensions

Main empirical constraints, mitigations, and next research steps



1 • Small listed-bank sample | Use bank clustered SEs, WCB p-values and LOBO/LOCO checks.

36 listed banks and slow-moving accounting variables.

Future extension: scale to a larger or more granular panel. Test finer tercile, quartile or decile splits.

2 • Customer deposits are only a broad proxy | Demand/Savings decomposition and common-sample bridge.

The ratio captures funding structure, not actual deposit rates, deposit betas, or customer switching costs.

Future extension: use bank-level deposit-rate data to estimate deposit betas directly.

3 • Loan share is a coarse asset-side proxy

It misses the fixed-versus-floating mix, repricing speed, securities duration and hedging of the loan book.

Future extension: separate floating and fixed loans, repricing gaps and hedging to trace the specific asset-side mechanism.

4 • Endogeneity and unobserved heterogeneity | Re-run H1 broad CustDep tests on composition sample

Banks that report decomposed deposits may be naturally more advanced; bank and month FE do not capture all non-slope-movement variation.

Future extension: add an instrument or deposit-rate data.

5 • Market Power

Well-established channel for franchise deposits to benefit (Drechsler et al., 2017) .

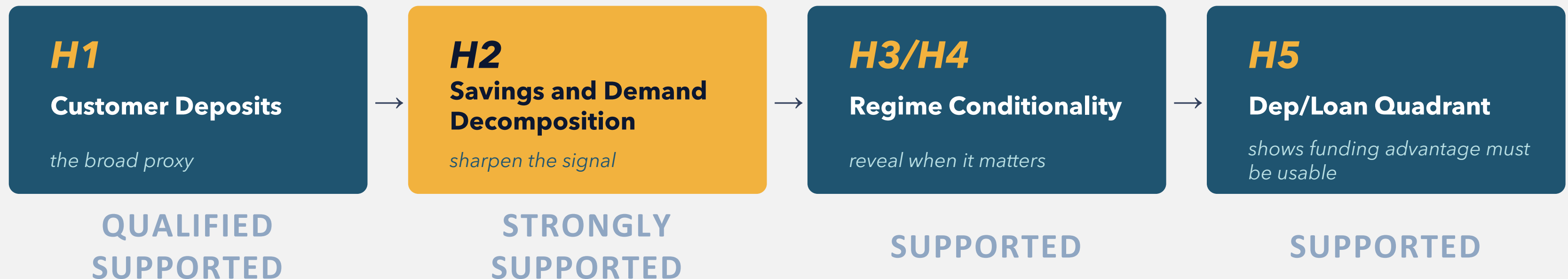
Future extension: Does market concentration amplifies the franchise-deposit channel. Can be tested via a triple interaction.

The Empirical Story

A nuanced conclusion



The markets price asymmetrically, differentiating deposit types conditional on the rate level, market stress and assets.



FOR PRACTITIONERS: A SCREEN DURING RATE-CYCLE TURNING POINTS

DEPOSIT COMPOSITION

HIGH RATES & MARKET STRESS

ASSET-SIDE CAPACITY



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